

Sayan Mondal

Solution Architect with 5+ years of enterprise technology experience spanning ServiceNow, agentic AI, SaaS transformation, application architecture, and AI/ML delivery. PhD researcher focused on causal cyber defense, trustworthy threat detection, and security knowledge graphs.

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PROFESSIONAL EXPERIENCE

Solution Architect

Mar 2025 - Present

Humanize Business Solution Pvt Ltd · Bengaluru, India

- Design solution architectures and agent workflows for enterprise discovery, documentation, specification, configuration, code generation, validation, and testing.
- Translate ServiceNow and SaaS transformation use cases into orchestrated workflows across Discover, Developer, and Decision agent capabilities.
- Define integrations with ITSM platforms, source repositories, version-control systems, and test-automation tools for near-touchless delivery.
- Apply RAG-based enterprise knowledge, AI guardrails, security controls, explainability, and traceability from discovery to production.

Software Engineer

Nov 2020 - Feb 2025

Virtusa · Hyderabad, India

- Developed and supported internal R&D applications using Python, ServiceNow, and AI/ML technologies.
- Managed ServiceNow instances and contributed to RFP solution design across the company portfolio.
- Designed application architectures and supported delivery from requirements through implementation and deployment.
- Provided technical training and mentoring and integrated AI/ML capabilities into enterprise workflows.

EDUCATION

PhD, Computer Science & Engineering2025 - 2029

IIIT Guwahati · BRAIN Lab

In progress · Causal cyber defense, active inference, security knowledge graphs, and trustworthy threat detection.

M.Tech, Software Engineering2022 - 2024

BITS Pilani

B.Tech, Computer Science & Engineering2016 - 2020

Techno India University, West Bengal

CORE EXPERTISE

Solution Architecture	ServiceNow
Agentic AI	Generative AI
Machine Learning	RAG
AI Guardrails	Cybersecurity
Enterprise Integrations	Python
Technical Strategy	Mentoring

CERTIFICATIONS

- ServiceNow Certified Application Developer
- ServiceNow Certified System Administrator
- Partner Pre-Sales Specialization: Now Platform
- Ethics in the Age of Generative AI

SELECTED ARCHITECTURE WORK

Agentic AI Delivery Workflow for Enterprise Software

Designed a reusable six-stage architecture spanning discovery, specification, configuration, development, validation, and testing. The pattern combines specialized agents, RAG-grounded artifacts, scoped access, validation controls, audit trails, and human approval gates.

Agentic AI · RAG · ServiceNow · Workflow orchestration · Human-in-the-loop controls

ServiceNow Solution Design & R&D Enablement

Translated business requirements into platform workflows, application structures, integration plans, and implementation-ready documentation while contributing to RFP solution design, reusable patterns, and team mentoring.

ServiceNow · ITSM · Python · AI/ML · Enterprise delivery · RFP solutioning

Enterprise Architecture Portfolio

Created anonymized, vendor-neutral reference designs for API-led integration, governed enterprise RAG, ServiceNow workflow modernization, secure cloud foundations, multi-agent orchestration, and AI observability and governance.

Reference architectures · Integration · Secure cloud · AI governance · Observability

[View the architecture portfolio](#)

SELECTED PUBLICATIONS

2026	SiNMULI: Novel Signed Network Approach for Malicious URL Identification arXiv preprint · Reported 99.89% accuracy and 99.80% F1 · arXiv:2608.19190
2026	CAIRN: Causal Active Inference for Real-Time Cyber Attack Diagnosis and Response ACM AsiaCCS '26 · Poster · DOI 10.1145/3779208.3804893
2026	EntroURL-Bench: A Feature-Rich Benchmark Dataset for Trustworthy Malicious URL Detection ISEC 2026 · Research Papers Track · Short paper · DOI 10.1145/3796563.3796574
2020	Global Positioning System Based Automated Railway Level Crossing ICCECE 2020 · DOI 10.1109/ICCECE48148.2020.9223092

Six publications in total. Full list: [sayanmondal2098.github.io/research/](#)

Research profiles: [ORCID](#) · [Google Scholar](#) · [ResearchGate](#)